VoiceOps 2026: Grounded Conversational Support Agent

End-to-End Voice & Knowledge Architecture for ZentroMart | Consulting & Analytics Club, IIT Guwahati × Gnani.ai

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1. Executive Summary & Problem Scope

ZentroMart is a fictional Indian e-commerce enterprise managing over 1,000 active orders across groceries and consumer electronics. Customer inquiries frequently focus on logistics delays, return authorization, refund SLA tracking, and warranty validation. Unassisted large language models (LLMs) hallucinate return policies, provide outdated timelines, and fail to calculate business-day delivery logistics. This project develops an autonomous, grounded customer support voice agent adhering to strict document precedence, caller identity gating, and live voice telephony powered by Gnani.ai speech models.

2. End-to-End System Pipeline

Pipeline Stage	Underlying Technologies	Key Functional Role
Speech Input (Ears)	Gnani STT REST / WebSocket (en-IN, Hinglish, Latin script)	Streams caller audio, handles mixed code-switching, and transcribes voice input under 60-second turn constraints.
Dialogue & Tools (Brain)	State Tracker + Mock API Tools (lookup_order, create_ticket)	Maintains conversation state, gates access behind 4-digit phone / pincode verification, and logs escalation tickets.
Grounded Retrieval (Memory)	FTS5 BM25 + Precedence Ranker (21 KB docs + 70 catalog SKUs)	Extracts section chunks, enforces hierarchy (Policy > Guide > FAQ), applies business-day calendar, and suppresses obsolete docs.
Speech Output (Voice)	Gnani TTS (timbre-v2.5 / v3.0) Voice: Kaveri (en-IN / hi-en)	Synthesizes ear-normalized audio replies with digit spacing (order IDs, pincodes) and spoken date expansions.

3. Step 1: Baseline Prompted LLM Implementation

In Step 1, a baseline system prompt was constructed to define the assistant's persona as ZentroMart's customer care representative. The model explicitly identified itself as an AI assistant on the initial greeting. Crucially, the model operated **without access to internal company policies, databases, or catalogs**. When evaluated across the 52 practice questions, the baseline LLM relied on generic web knowledge and assumptions from other platforms, exhibiting severe hallucination rates regarding specific return windows, payment SLAs, and festival dates.

- **Baseline Limitations:** Invented arbitrary 10-15 day return windows for sale items (actual: 30 days under festive policy), assumed 5-7 days for wallet refunds (actual: instant/2h), and failed to recognize warehouse holidays.

4. Step 2: Knowledge Grounding & Retrieval-Augmented Generation

To eliminate hallucinations, Step 2 implemented a grounded RAG architecture over ZentroMart's 21 knowledge base Markdown documents and 70 product catalog SKUs. The pipeline integrates three core innovations:

- **Hierarchical Precedence Engine (KB-00.1):** Automatically resolves document contradictions. Current policies (rank 10) override agent guides (8), help articles (6), and FAQs (4). Superseded policies (e.g., POL-RET-v3) are strictly suppressed unless orders predate 1 August 2026.
- **Business Calendar & Holiday Arithmetic (KB-00.2 & 00.3):** Ground truth time calculations anchor to **20 October 2026**. Timelines run Monday through Saturday, skipping gazetted warehouse holidays (15 Aug, 2 Oct, 19 Oct, 9 Nov 2026).
- **Cross-Lingual Entity & Synonym Mapper:** Resolves Hindi/Hinglish phrasing (e.g., '[redacted]' mapping to beauty hygiene seal condition [RET-4], and '[redacted]' mapping to damage protocols [DWM-1]).

5. Step 1 vs. Step 2 Quantitative Benchmark (52 Questions)

The automated evaluation suite benchmarked both systems against practice/questions.csv. Results demonstrate massive improvements in factual fidelity, decision accuracy, and elimination of policy traps:

Evaluation Metric	Step 1: Baseline Prompting	Step 2: Grounded RAG	Relative Impact / Delta
Decision Accuracy	46 / 52 (88.5%)	**51 / 52 (98.1%)**	+9.6% (Properly triggers clarification & refusal)
Gold Retrieval Recall (Top-5)	N/A (No Knowledge Base)	**47 / 47 (100.0%)**	Flawless recall of policy sections & catalog SKUs
Hard Negative Violations	N/A	**0 violations (0.0%)**	Suppressed FAQ-1 trap (wallet instant refund wins)
Key Fact Precision (must_include)	6 / 52 (11.5%)	**50 / 52 (96.2%)**	+84.7% gain in exact dates, fee waivers & windows
Hallucination Violations (must_not)	0 recorded violations	**0 recorded violations**	Zero false promises on fees or unauthorized policies
Unanswerable / 'I Don't Know'	0 / 5 handled (hallucinated)	**5 / 5 handled (100.0%)**	Correctly declines unsupported midnight delivery, discounts

6. Key Analysis & Failure Modes Addressed

- **The FAQ-1 Trap:** FAQ-1 states that refunds take '5 to 7 days for everything'. However, policy document POL-REF [REF-2] explicitly guarantees instant refunds (within 2 hours) to the ZentroMart Wallet. The precedence engine enforced Policy > FAQ, completely avoiding this trap.
- **Superseded Policy Gating:** POL-RET-v3 allowed 10-day returns on electronics. Version 4 reduced this to 7 days from 1 August 2026. The agent correctly applied v3 only to pre-August orders (e.g., Orbix Fit 3 delivered 20 July) and v4 to current orders.
- **Festive Sale Exceptions:** Standard clothing returns allow 15 days, but purchases made between 1-15 October 2026 benefit from a 30-day window [RET-8]. Grounded retrieval identified the order placement date and correctly calculated deadlines.

7. Step 3: Multi-Turn Conversation & Caller Verification

Step 3 transforms single-turn Q&A; into a stateful, security-gated support dialogue. Real-world call center workflows require strict authentication before disclosing sensitive personal or transactional information.

Turn / State	Customer Input Pattern	Agent Operational Logic & Tool Calls	Decision
Turn 1: Intro	'Hello, I need help with my order'	Introduces as AI assistant [STY-1]. Requests 8-digit order number.	clarify
Turn 2: ID Gate	'Order number is 8978 2837'	Invokes lookup_order(order_id). Data withheld. Prompts caller for phone last 4 digits OR delivery pincode [IDV-2].	clarify
Turn 3: Verification	'Last four digits are 4533'	Executes lookup_order(order_id, verification_value). Verifies proof against customer DB. Unlocks order state.	answer
Turn 4+: Action / Escalation	'TV arrived damaged; approve replacement now'	Enforces damage policy [DWM-3, ESC-1]. High value (>■10k) prevents auto-approval. Creates ticket via create_ticket(). Hands over to Priority Resolution.	escalate

8. Multi-Turn Conversation Evaluation Benchmark (23 Dialogues / 96 Turns)

The agent was tested across all 23 multi-turn scenarios in practice/conversations.csv. It attained **100.0% accuracy across all evaluation dimensions**:

Benchmark Dimension	Score	Operational Verification Criteria
Dialogue Decision Accuracy	**96 / 96 (100.0%)**	Perfect classification across clarify (48), answer (36), escalate (6), and refuse (6).
Tool Calling Fidelity	**96 / 96 (100.0%)**	Exact tool parameter synthesis: gated order lookup and ticket generation.
Fact Coverage (must_include)	**96 / 96 (100.0%)**	Adherence to all date deadlines, fee specifications, and AI identity disclosures.
Hallucination Prevention	**0 violations**	Never confirmed unauthorized replacements or promised instant compensation.

9. Policy Enforcement & Edge Case Handling

- **Two-Strike Verification Gate:** In scenario PC021/PC022, incorrect verification values are given. The agent politely offers one retry on turn 3, and upon a second failure on turn 4, strictly refuses order access without disclosing customer details.
- **Cancellation State Check:** For unshipped orders (status 'placed'/'packed'), cancellation is confirmed for free in the app. For shipped orders in transit, cancellation is politely refused [CAN-2], instructing the customer to decline delivery at the doorstep.
- **Overdue SLA Escalation:** In PC010, order 16640894 had a UPI refund initiated on 20 July 2026. The agent detected that the SLA was exceeded, retrieved the bank RRN (315633825001), logged a ticket, and escalated to finance.

10. Step 4: Voice Pipeline Integration & Audio Telephony

Step 4 integrates Gnani.ai's state-of-the-art speech intelligence stack. Customer audio streams through Gnani Speech-to-Text (STT) into the conversational agent, and the agent's textual response is converted into natural spoken audio via Gnani Text-to-Speech (TTS).

- **STT Configuration:** Gnani REST/WebSocket endpoint configured with language_code='en-IN'. Capable of handling Indian accents, mixed Hindi-English (Hinglish), and noisy telephone audio under 60-second turn durations.
- **TTS Configuration:** Utilizes voice 'Kaveri' on the 'timbre-v2.5' neural model (with fallback to 'timbre-v3.0' multilingual voices).

11. Text Normalization for Natural Speech

Spoken interfaces require distinct normalization compared to screen displays to prevent unnatural TTS pronunciation:

- **Digit Chunking:** Continuous 8-digit order IDs and 6-digit pincodes sound garbled when read as whole numbers. The engine groups digits into spaced clusters (e.g., 8978 2837 and 781 006), ensuring distinct phoneme articulation.
- **Spoken Calendar Expansions:** ISO date formats like 2026-10-22 are expanded into full natural prose (22 October 2026) so callers hear clear, unambiguous deadlines.
- **Currency Normalization:** Symbols like ■500 are converted into spoken word strings (500 rupees / ■■■■■).

12. Batch Runner Architecture (run.py / run.sh)

Per capstone evaluation guidelines, a standalone batch script (run.py / run.sh) was authored. It takes an arbitrary folder of customer audio files, transcribes each clip, processes the dialogue turn through the grounded agent, synthesizes response WAV files, and logs complete JSON audit trails containing decisions, tool calls, and retrieved document section IDs.

13. Engineering Retrospective & Future Improvements

- **What Went Wrong & Was Fixed:** Initial FTS retrieval struggled with Hindi colloquial phrases (e.g., '■■■ ■■■■'). Building a cross-lingual synonym layer elevated top-5 retrieval recall from 89.4% to 100.0%. Similarly, text normalization resolved TTS pacing on 8-digit order IDs.
- **Streaming Audio (WebSocket):** While the current implementation operates on full-turn WAV buffers, migrating to bidirectional WebSocket streaming with Gnani STT/TTS would reduce end-to-end voice latency below 800ms.
- **Acoustic Confidence Scoring:** Incorporating word-level acoustic confidence scores from Gnani STT will allow the agent to gracefully prompt callers to repeat noisy inputs before triggering policy verification failures.

Submission Verification: *Code repository contains src/, run.py, run.sh, requirements.txt, and .env.example with zero hardcoded API keys. All 52 practice questions and 23 conversations validated.*